PCA Face Reconstruction on the Olivetti Dataset: Quality and Computational Efficiency

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***Abstract*—Principal component analysis (PCA) provides compact face representations, but selecting a practical configuration requires assessing reconstruction quality and computational cost together. This study evaluates a PCA pipeline on the Olivetti dataset of 400 grayscale images with 4096 pixels each. Covariance eigendecomposition and economy singular value decomposition (SVD) are compared using reconstruction, structural, classification, and resource metrics. With $k = 20$ components, PCA retains 76.34% of the variance and reduces mean squared error from 0.01927 to 0.00456. The structural similarity index reaches 0.698, while self-excluded nearest-neighbor classification achieves 94.8% accuracy using a basis fitted to all images. The within-/between-subject distance ratio decreases from 0.675 in pixel space to 0.562 in PCA space. A fixed 280/120 split yields test error 0.00538 and accuracy 89.2% on unseen images of the same subjects. Economy SVD produces numerically equivalent reconstructions while reducing median decomposition time from 23.4 to 0.97 s on the tested machine. Coefficient storage requires 64 kB, excluding the shared basis and mean, compared with 13.1 MB for the original float64 data. These results characterize the quality–cost trade-off of an established linear baseline and show why component selection should reflect the intended task.**

***Index Terms*—Eigenfaces, face reconstruction, principal component analysis, singular value decomposition, structural similarity.**

I. Introduction

A grayscale face image with 64×64 pixels contains $d = 4096$ values, many of which are correlated. A compact representation can reduce storage and processing costs while preserving information useful for reconstruction and identity classification. Pixel-space distances also reflect changes in illumination, expression, and pose. Reconstruction error alone therefore provides an incomplete assessment of a face representation.

PCA finds orthogonal directions of maximal variance and reconstructs images from the retained components [1]. Its eigenface interpretation connects dimensionality reduction with appearance-based face representation [2]. Choosing a component count and choosing a solver address different requirements: the former controls retained information, whereas the latter determines the cost of estimating the same subspace.

This study examines both choices on the Olivetti face database [3], [4]. A from-scratch implementation compares covariance eigendecomposition with economy SVD and checks their numerical agreement. Reconstruction is assessed using mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity (SSIM). Identity information is assessed using one-nearest-neighbor (1-NN) classification and within-/between-subject distances. Runtime measurements complement complexity and storage estimates, while a held-out split evaluates images excluded from basis estimation. Together, these analyses relate component and solver choices to the quality and cost of this linear baseline.

II. Methods

A. PCA Formulation

Let $X \in \mathbb{R}^{n \times d}$ hold one face per row and $\mu = \frac{1}{n}\sum_i x_i$ be the mean face. After centering, $\tilde{X} = X - \mathbf{1}\mu^\top$, PCA solves

$$C v_j = \lambda_j v_j, \qquad C = \frac{1}{n-1}\tilde{X}^\top \tilde{X}, \tag{1}$$

with eigenvalues sorted in descending order. Keeping the leading k eigenvectors as $W = [v_1, \ldots, v_k] \in \mathbb{R}^{d \times k}$ gives the code and the linear reconstruction

$$z = W^\top (x - \mu), \qquad \hat{x} = \mu + Wz = \mu + \sum_{j=1}^{k} z_j v_j. \tag{2}$$

The explained-variance ratio is $\eta_j = \lambda_j / \sum_{i=1}^{d} \lambda_i$ [5]. The denominator includes the full spectrum; normalizing only over retained components would overstate variance retention. Centering limits the rank of $\tilde{X}$ to at most $n-1$.

Each eigenvector, reshaped to 64×64, is an *eigenface*: the leading components show broad intensity patterns, while later components show more localized variation (Fig. 2). The complete pipeline is summarized in Fig. 1.

B. Two Equivalent Solvers and Their Cost

Covariance route. Form $C \in \mathbb{R}^{d \times d}$ explicitly and eigendecompose it, (1). Forming C costs $O(nd^2)$ and its eigendecomposition $O(d^3)$; the matrix itself occupies d^2 doubles. *SVD route.* Decompose the centered data directly, $\tilde{X} = U\Sigma V^\top$. Since $\tilde{X}^\top \tilde{X} = V\Sigma^2 V^\top$, the columns of V are the eigenvectors in (1) and

$$\lambda_j = \sigma_j^2 / (n-1), \tag{3}$$

so the two routes are algebraically identical, while the economy SVD costs $O(nd\min(n,d))$ and never materializes C [6]. A third variant, the Gram-matrix trick ($\tilde{X}\tilde{X}^\top \in \mathbb{R}^{n \times n}$, eigendecompose, recover $V = \tilde{X}^\top U \Sigma^{-1}$), costs $O(n^2 d + n^3)$ when restricted to nonzero singular values. This variant is outside the timing comparison. Linear variants such as two-dimensional PCA [7] and nonlinear kernel PCA [8] extend the variance-ordering principle; the present comparison focuses on the standard linear formulation.

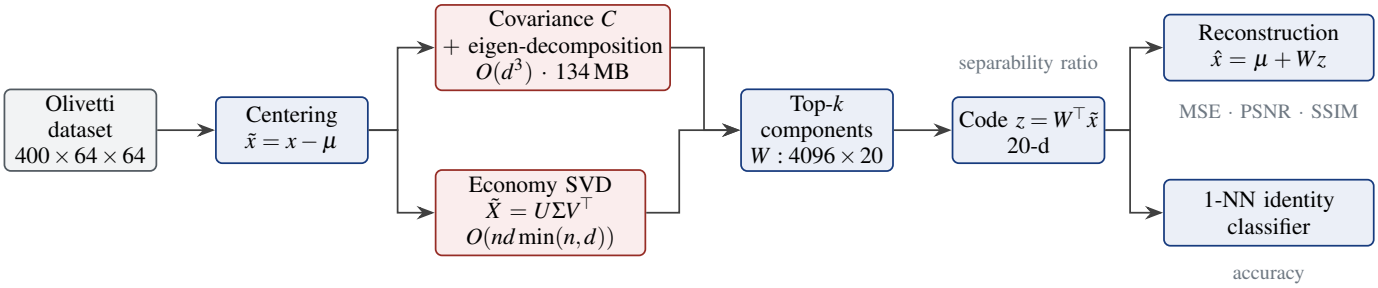


Fig. 1. PCA pipeline at $k = 20$. The two decomposition routes estimate the same principal subspace. The covariance storage label assumes float64; runtime and array storage are compared in Table II.

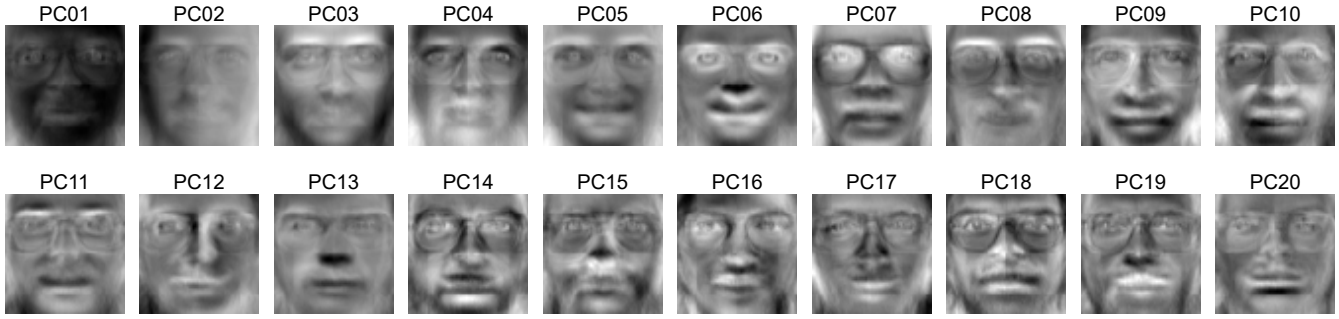


Fig. 2. The first 20 eigenfaces (PC01–PC20, in descending order of eigenvalue). Each component is displayed with its own symmetric intensity scale to reveal spatial patterns; sign and display contrast do not indicate component importance.

C. Evaluation Metrics

Reconstruction. For image i, $\mathrm{MSE}_i = d^{-1}\|x_i - \hat{x}_i\|_2^2$ and $\mathrm{PSNR}_i = 10\log_{10}(1/\mathrm{MSE}_i)$ for unit-range pixels. Both metrics are averaged over images, so mean PSNR is not computed from the aggregate MSE. Structural similarity [9] is

$$\mathrm{SSIM}(x,\hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + c_1)(2\sigma_{x\hat{x}} + c_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + c_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + c_2)}. \tag{4}$$

Here μ_x, σ_x^2, and $\sigma_{x\hat{x}}$ denote local means, variances, and covariance, and c_1, c_2 stabilize the ratios. We use scikit-image's `structural_similarity` function with unit data range and otherwise retain its default settings, then average the image scores.

Identity information. Euclidean 1-NN classification is evaluated on 40 identities without tuning. In the full-data evaluation, PCA is fitted once to all images and each query is excluded from its own neighbor search. This is a self-excluded evaluation with a fixed basis, not cross-validation that refits PCA for every query. Separability is summarized by the ratio of mean within-subject to mean between-subject distances; a lower ratio means smaller within-subject distances relative to between-subject distances.

Resources. Decomposition time is the median of three wall-clock runs on one machine. Complexity, array storage, and coefficient storage are analyzed separately. Array sizes assume eight bytes per element and are reference float64 estimates, not measured process peak memory.

III. Experimental Setup

The scikit-learn version of the Olivetti face dataset [3], [4] contains $n = 400$ grayscale images of 64×64 pixels from 40 subjects (ten images each), with pixel values normalized to $[0, 1]$. The pipeline of Fig. 1 is implemented in Python with NumPy and Matplotlib, with the random seed fixed at zero so that the sampled image (index 172, subject 17) and all figures are reproducible. Unless stated otherwise, $k = 20$ matches the course requirement; sweeps use $k \in \{5, 10, 20, 50, 100\}$ for quality and $\{5, 10, 20, 30, 50, 66, 100, 150\}$ for accuracy. Both solvers are implemented and cross-checked: the reconstructions agree to a maximum absolute difference of 1.97×10^{-7}, and the reported explained variance agrees. The original timing routine uses centered float32 input; float64 storage estimates are reported independently of that benchmark. For the held-out check, the first seven images of each subject form the training set and the last three form the test set. The training mean and basis are estimated without test images, and training projections serve as the 1-NN gallery.

IV. Results

A. Reconstruction Quality

Fig. 3(a) and Table I summarize the quality–complexity trade-off. Relative to the mean-face baseline (0.01927), $k = 20$ reduces the MSE to 0.00456 (a 76.3% reduction), with mean PSNR 23.640 dB. Per-image MSE is 0.00456 $\pm$ 0.00150 (mean $\pm$ standard deviation). SSIM increases from 0.451 for the mean-face baseline to 0.698 at $k = 20$.

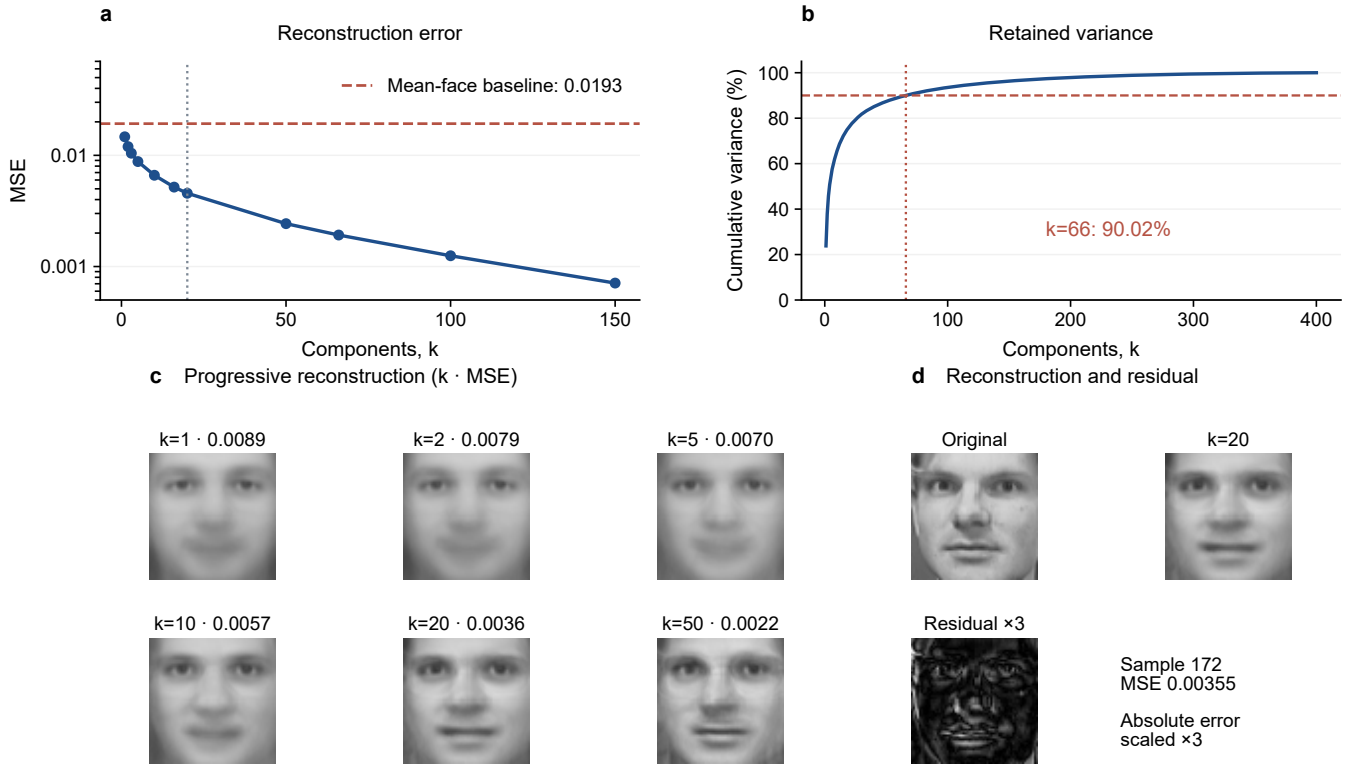


Fig. 3. Effectiveness at a glance. (a) Reconstruction error versus k (log scale) with the mean-face baseline and the $k = 20$ operating point; (b) full-spectrum cumulative explained variance, marking $k = 66$ where 90% is reached; (c) cumulative reconstructions of the sampled face for $k = 1, 2, 5, 10, 20, 50$ with per-panel MSE; (d) the same face, its $k = 20$ reconstruction and the $3\times$ amplified residual, which concentrates on edges and high-frequency structures.

TABLE I
FULL-SAMPLE RECONSTRUCTION QUALITY. PSNR AND SSIM ARE AVERAGED OVER IMAGES; $k = 0$ USES THE MEAN FACE.

k	MSE	PSNR (dB)	SSIM
0 (mean face)	0.01927	17.516	0.451
5	0.00879	20.798	0.540
10	0.00662	22.036	0.615
20	**0.00456**	**23.640**	**0.698**
50	0.00243	26.359	0.812
100	0.00125	29.182	0.888

The discarded eigenvalues determine the full-data reconstruction error:

$$\mathrm{MSE}_k = \frac{\|\tilde{X} - \tilde{X}WW^\top\|_F^2}{nd} = \frac{n-1}{nd}\sum_{j>k}\lambda_j. \tag{5}$$

The factor $(n-1)/n$ reconciles sample-covariance normalization with per-pixel MSE. The original log reports the uncorrected value $\sum_{j>k}\lambda_j/d = 0.004571$ versus observed 0.004559. Its 0.25% relative difference is accounted for by this normalization factor, rather than a solver discrepancy.

Reaching 90% variance retention requires $k = 66$, compared with 76.34% at $k = 20$. The cumulative reconstructions in Fig. 3(c) show the recovery of finer facial structure as k increases. For the illustrated image, the residual in Fig. 3(d) is prominent near facial boundaries, glasses, and the hairline, indicating detail lost at the selected component count.

B. Identity Separability and Classification

Fig. 4(a) and (b) compare the within- and between-subject distance distributions in pixel space and in the k-dimensional PCA space. In pixel space the two distributions overlap heavily: the within-subject mean distance (8.3777) reaches 0.675 of the between-subject mean (12.4144). PCA reduces within-subject distances more than between-subject distances (means 6.0258 and 10.7313), lowering the ratio to 0.562. Self-excluded 1-NN accuracy reaches 94.8% at $k = 20$ (Fig. 4(d)) and ranges from 95.2% to 95.5% at the evaluated $k \geq 50$. These are within-dataset results with a basis fitted to all images; Section IV-D reports the separate held-out gallery protocol.

C. Runtime and Storage

Table II and Fig. 5 compare the two solvers on the same machine with the same data. The economy SVD needs 0.97 s (median of three runs; range 0.82–1.43 s) against 23.4 s (range 21.4–24.7 s) for the covariance route, a $24\times$ speed-up; a second independent run of the same pipeline reproduced the ordering (1.22 s versus 20.4 s), although absolute times vary between runs. The gap is explained by the operation counts: with $n \ll d$ the covariance route spends $O(d^3)$ on a $d \times d$ eigendecomposition, while the economy SVD works with the $n \times d$ data and costs $O(nd\min(n,d))$. For these dimensions, the leading-order ratio $(d/n)^2 \approx 105$ illustrates the scale difference, with algorithm-dependent constants omitted. The covariance matrix alone would occupy 134 MB in float64,

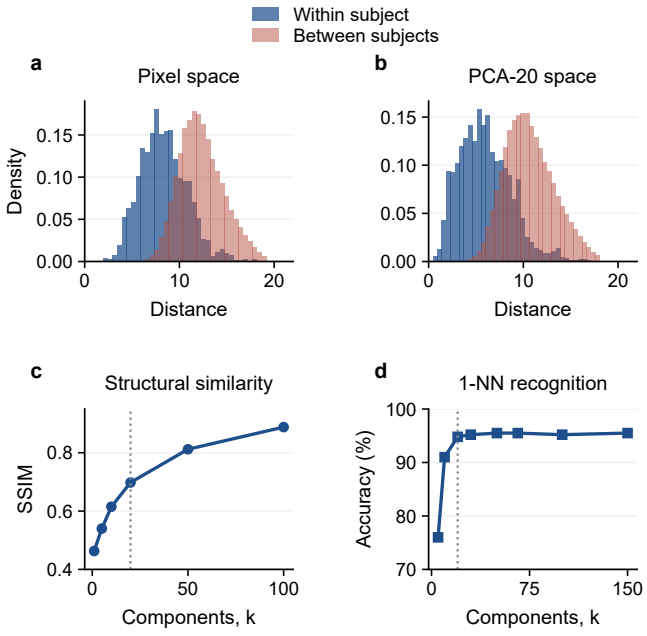


Fig. 4. Separability and task metrics. (a) Within- and between-subject distance distributions in pixel space; (b) the same in PCA-20 space; (c) mean SSIM versus k; (d) self-excluded 1-NN accuracy using the full-data basis. Dashed vertical lines mark $k = 20$.

TABLE II
SOLVER TIMES: MEDIAN AND RANGE OF THREE RUNS. ARRAY STORAGE ESTIMATES ASSUME FLOAT64 AND EXCLUDE WORKSPACE.

Solver	Median (s)	Range (s)	Array (MB)
Covariance + eigh	23.4	21.4–24.7	134
Economy SVD	0.97	0.82–1.43	13.1

Arrays: covariance matrix C (4096×4096) and centered data $\tilde{X}$ (400×4096), respectively. These are reference array sizes, not measured peak memory.

compared with 13.1 MB for the data matrix. These estimates exclude solver workspaces and outputs.

Encoding and reconstruction require approximately $4kd = 0.33$ million floating-point operations per image when multiplication and addition each count as one operation. The recorded median latency is 0.03 ms per image over five passes through the dataset. At $k = 20$, the coefficients require 160 bytes per image in float64, or 64 kB for all images, approximately $205\times$ less than the original pixel array. Decoding also requires the shared basis and mean. Including these arrays gives $8(nk + dk + d) = 752{,}128$ bytes, approximately 0.752 MB, for the model and all codes. Thus the coefficient-only reduction should be distinguished from the complete representation's storage cost.

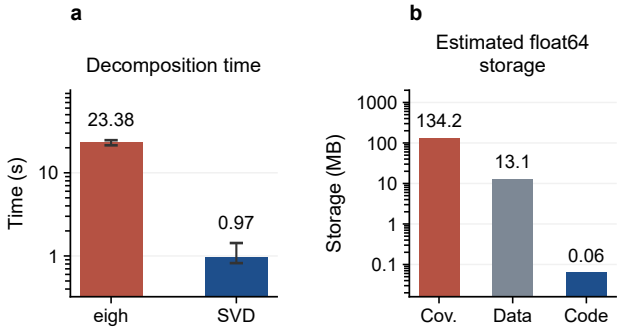


Fig. 5. Efficiency at a glance: decomposition time on a log scale with the observed range over three runs (left); reference float64 array storage for the covariance matrix, centered data, and PCA codes (right).

D. Held-out Evaluation

The fixed 280/120 split evaluates $k = 20$ on images excluded from basis estimation. Test reconstruction MSE rises from 0.00456 to 0.00538 (+17.9% relative to the in-sample value), PSNR falls from 23.640 to 22.962 dB and SSIM from 0.698 to 0.654; 1-NN accuracy with the training gallery drops from 94.8% to 89.2%, while the within-/between-subject distance ratio is similar (0.562 for the full-data projections versus 0.559 for held-out projections). All test subjects are represented in the training gallery. This performance gap combines changes in basis estimation and gallery size; a single split at one k does not establish generalization to new identities or validate the full component-count sweep.

V. Discussion

Reconstruction and classification favor different component counts. Increasing k continues to improve pixel error and SSIM, whereas self-excluded 1-NN accuracy approaches a plateau near $k = 50$. The operating point should therefore reflect the target metric and the marginal benefit of additional components.

The solver comparison separates representation quality from computational cost. The two solvers estimate the same leading subspace and produce numerically equivalent reconstructions in this experiment. The runtime advantage of economy SVD is consistent with its lower complexity when $n \ll d$, although the measured speed-up is specific to the implementations and machine tested.

The lower distance ratio suggests that the retained subspace preserves relatively more between-subject variation than within-subject variation. It does not identify the acquisition factors responsible for that change. The held-out results further restrict the supported classification claim to new images of known identities under the tested gallery protocol.

VI. Limitations and Future Work

Evidence is limited to one small dataset and one fixed split. Repeated splits with PCA refitted within each training fold would quantify partition sensitivity; cross-session recordings and unseen identities would test broader recognition claims.

Timings lack a complete hardware and numerical-library configuration for cross-machine comparison. Array storage estimates exclude solver workspaces. Separability pools illumination and expression within subjects; factor-specific analysis and comparisons with 2-D PCA or kernel PCA could clarify which variations the linear representation retains.

VII. Conclusion

This study jointly evaluates reconstruction quality and computational cost for PCA face reconstruction. With $k = 20$ components, PCA retains 76.34% of the variance and reduces reconstruction MSE by 76.3% relative to the mean-face baseline. Economy SVD produces numerically equivalent reconstructions while reducing median decomposition time from 23.4 to 0.97 s. The results support SVD for this $n \ll d$ setting and show that component selection should reflect the reconstruction or classification objective. Held-out findings concern new images of known subjects. The accompanying code, figures, and recorded experiment output support reproduction of the analysis.

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